
METHODS & THEORY

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$\Phi_1 \rightarrow \Phi_2 \rightarrow \Phi_3 \cdot \text{excess}^3 \cdot \text{RECD}$

Integrating Φ_3 (excess³) into RECD Dynamics

From Parallel Metric to
Structural Clock Contribution

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Abstract

The Systemic Tau (τ_s) framework and the Discrete Extramental Clock (RECD) identify critical transitions through ordinal relational reorganization rather than through growth in univariate magnitude. A three-level hierarchy of ordinal conjunctions— Φ_1 (coincidence), Φ_2 (persistent relation), and Φ_3 (irreducible synergistic surplus, operationalized by the continuous proxy excess³)—has been formalized and validated on synthetic systems. This work bridges the gap between the nested theoretical hierarchy and the legacy operational pipelines used in epidemiological and cardiac surveillance. In these pipelines, excess³ has typically remained a parallel diagnostic: it contributes to early-warning scores and nested mass fractions but does not systematically enter the discrete-time update.

We address it in two complementary ways. Proposal 1 multiplies the existing RECD increment by a surplus-gated gain Γ_3 : the change is minimal, so that any modification of the clock can be attributed to Φ_3 by setting $\beta = 0$. Proposal 2 rebuilds the advance as two additive channels and lets Φ_3 also deepen Feigenbaum compression: it is more expressive, but harder to audit in the field, and is therefore treated as experimental. Both proposals share the same hardened activation gate A_3 , because the main practical risk is not “too little Φ_3 ” but trivial activation whenever the system sits in the chaos band for long stretches (hyper-persistence).

The analysis proceeds as follows: (i) diagnose why parallel deployment of Φ_3 leaves the emergence hierarchy dynamically incomplete; (ii) define Proposal 1 and Proposal 2; (iii) specify activation rules that require surplus change, residual synergy, and dual-scale persistence; (iv) validate on synthetic suites S0–S2 with pre-specified metrics (A_3 rates, T_{final} , ρ_3 , $\text{corr}(\Gamma_3 - 1, f_3)$, noise robustness); and (v) run paired empirical analyses on SDDB/NSRDB Holter recordings and DengAI San Juan/Iquitos series—two domains that share chronic band-C occupancy but differ in timescale and proxy structure. Interpreted jointly, the hardened A_3 gate controls hyper-persistence in both domains; legacy ΔT advances only when chaos-band occupancy permits movement of Δt^{base} ; and event- and outbreak-level specificity under fixed thresholds remains to be demonstrated. Proposal 1 is the operational default; Proposal 2 is retained for synthetic structural comparison.

excess³ is retained throughout as a pre-specified hybrid proxy (0.6 Syn + 0.4 Surp) rather than as a complete partial-information decomposition. Causal claims are restricted to dynamical influence on the discrete clock index.

1. Structural incompleteness of parallel Level-3 deployment

1.1 Operational synthesis of the 2026 framework

The present analysis sits on the Systemic Tau / RECD framework developed in prior preprints [1, 5], with excess³ as the continuous Level-3 proxy [2]. Let $X(t) \in \mathbb{R}^d$ denote a multivariate series with $d \geq 2$. In successive windows of length W , Systemic Tau is the mean pairwise Kendall rank correlation [8]:

$$\tau_s(k) = \frac{2}{N(N-1)} \sum_{i < j} \tau_K(r_i^{(k)}, r_j^{(k)}) \in [-1, 1].$$

Operational thresholds are motivated by the asymptotic variance of Kendall’s τ under independence and by Feigenbaum universality [6, 7] ($\delta \approx 4.6692016091$):

Regime	Condition	Operational interpretation
S	$\tau_s \geq +0.50$	Strong ordinal coherence
C	$ \tau_s < 0.41$	Chaos band / elevated ordinal volatility
A	$\tau_s \leq -0.41$	Strong antisynchronization

Equation (1) and equation (2) answer different questions [1]. Equation (1) is the operational clock used in dengue, Holter, and RQA pipelines [3, 4, 13]: it advances t_k from τ_s , a gate, and a state factor. Equation (2) is the theoretical mass decomposition of the conjunction hierarchy across Φ_1 – Φ_3 [1]. Nothing forces them to agree. In practice they often do not: f_3 can be large in (2) while (1) never multiplies or adds a Φ_3 term. That mismatch—not a preference for one formalism—is the starting point of the present analysis.

	Legacy equation (1)	Nested equation (2)
Role	Operational pipeline (epidemiology, RQA, cardiology)	Theoretical level-mass representation
Question it answers	How does discrete time advance in the field?	How is the increment partitioned across Φ_1 – Φ_3 ?
Increment	$\Delta t_k \cdot g(\tau_s) \cdot \alpha(\text{state})$	$\sum_{\ell} \alpha_{\ell}(\lambda) \Phi_{\ell}^*$
Role of Φ_3	Absent from Δt_k (or present only as an adjunct)	As mass f_3 , with λ typically constructed from τ_s alone
Integration path	Modulation of (1) by Φ_3 via Γ_3 or a dual channel	Compatible; f_3 reported in parallel

The classical RECD construction advances cumulative discrete time from a base interval Δt_k , a gate $g(\tau_s)$, and a state factor $\alpha(\text{state})$:

$$t_{k+1} = t_k + \Delta t_k \cdot g(\tau_s(k)) \cdot \alpha(\text{state}_k). \quad (1)$$

In the nested theoretical formulation, the increment is a weighted sum of conjunctions:

$$\Delta \text{RECD}(t) = \alpha_1(\lambda) \Phi_1(t) + \alpha_2(\lambda) \Phi_2(t) + \alpha_3(\lambda) \Phi_3^*(t), \quad (2)$$

where Φ_3^* denotes the binary indicator Φ_3 or the continuous proxy excess³, and λ is a regime cue (for example, a transform of $|\tau_s|$).

Replacing equation (1) with equation (2) everywhere is not viable for applications: field pipelines, historical comparisons, and RQA couplings are already written against (1). A wholesale replacement would break continuity with published dengue and Holter series and would mix a change of theory with a change of implementation. Proposal 1 therefore does not replace equation (2): it defines a map F acting on equation (1) and conditioned on surplus evidence, so that the operational clock and nested mass fractions can be compared side by side. Equation (2) remains the diagnostic mass decomposition; equation (1) remains the deployed clock, now optionally Φ_3 -aware.

excess³ is defined a priori as

$$\text{excess}^3(t) = 0.6 \cdot \text{Syn}(t) + 0.4 \cdot \text{Surp}(t), \quad (3)$$

where Syn denotes the multiinformation residual relative to a pairwise baseline and Surp denotes the log-ratio surprise under independence of ordinal symbols (Bandt–Pompe permutation entropy [9]). The hybrid weights 0.6/0.4 and the continuous proxy construction follow the pre-specified excess³ protocol [2]; they are not a complete partial-information decomposition in the sense of Williams and Beer [12]. The weights are fixed and are not re-estimated on the target dataset.

1.2 Parallel versus dynamical deployment of Φ_3

Three distinct roles of Level 3 appear in the 2026 framework [1, 2]:

1. Parallel early-warning statistic. Δexcess^3 between a baseline window and an approach window (for example, pre-ventricular-fibrillation intervals in SDDb [4, 14]; synthetic G0–G3 contrasts), in the broader early-warning tradition of critical-transition indicators [10].
2. Mass / increment fraction in equation (2): $f_3 = \alpha_3 \Phi_3^* / \sum_{\ell} \alpha_{\ell} \Phi_{\ell}^*$.
3. Structural contribution to the operational interval Δt_k in equation (1) and to hyper-persistence pipelines (RQA, six-mode consensus) [3, 13].

Roles (1) and (2) are comparatively well developed. Role (3) remains without a single, unified specification:

- In epidemiological surveillance and RQA refinement, the clock and associated signals are constructed from τ_s , persistence $P(k)$, $g(\tau_s)$, LAM/TT/DET, and states C/D/T. excess³ enters as an additional mode or adjunct metric, not as a canonical modulator of Δt_k .
- In equation (2), $\alpha_3(\lambda)$ elevates Level-3 mass when λ indicates chaos, yet λ is typically constructed from τ_s (or from a known control parameter r). Consequently, the weight of Φ_3 depends on the τ_s regime rather than on the observed irreducibility of excess³ within that window. A system may occupy band C with low excess³ (pair-dominated chaos) or with high excess³ (chaos carrying order- ≥ 3 surplus). These regimes warrant distinct clocks if the conjunction hierarchy is to govern the discrete-time construction.
- Without a $\Phi_3 \rightarrow$ clock channel, the hierarchy $\Phi_1 \subset \Phi_2 \subset \Phi_3$ remains descriptive in terms of mass and predictive as an early-warning signal, but is not generative of the discrete time index used in operational pipelines.

1.3 Dynamical closure of the hierarchy

Definition (dynamical closure). The conjunction hierarchy is dynamically closed if there exists a family of implementable maps F such that

$$\Delta t_k = F(\Delta t_k^{\text{base}}, g(\tau_s), \alpha(\text{state}), \Phi_1, \Phi_2, \text{excess}^3; \theta)$$

with pre-specified parameters θ , and if there exist falsifiable predictions on T_{RECD} , lead times, and mass fractions that fail when the Φ_3 channel is ablated (F independent of excess³).

Incorporation of Φ_3 into F is required so that:

1. Ordinal depth and the clock remain distinguishable. Without Φ_3 in F , two trajectories sharing $\tau_s(k)$ and $P(k)$ yield identical clocks even when their order-3 surplus differs.
2. Marginal contribution is testable. Comparison of RECD with and without the Φ_3 channel constitutes a controlled ablation experiment.
3. Nested theory (2) and pipeline (1) are aligned. The two coexisting formalisms are thereby unified without reformulating the RECD construction from its foundations.

This requirement does not imply that excess³ constitutes a complete partial-information atom, nor that the discrete clock represents physical time. It requires only that the index t_k respond to the structural layer that the framework treats as deepest.

1.4 Design constraints inherited from the framework

Any admissible construction must:

- preserve the thresholds $\tau_{\text{ch}} = 0.41$, $\tau_{\text{st}} = 0.50$, and the constant δ ;
- leave the 0.6/0.4 excess³ weights unaltered on target data;
- remain computable for short windows ($W \sim 13$) and moderate dimension d ;
- admit phase-shuffle nulls that destroy cross-dependence while preserving marginal spectra;
- avoid trivial activation of Φ_3 throughout band C (tropical hyper-persistence: 69–87% of time in the hyper-persistent core);
- remain compatible with RQA as a signal refinement rather than as a substitute for the clock.

1.5 Two proposals rather than one

A single map from Φ_3 into t_k is not uniquely determined by the framework. Two design pressures pull in opposite directions:

1. Minimal change and auditability. Surveillance pipelines require a drop-in factor on the clock already in use, with a one-parameter off-switch ($\beta = 0$) that recovers baseline behaviour exactly. That pressure yields Proposal 1: multiply the existing increment by $\Gamma_3 \geq 1$.
2. Structural fidelity to nested mass. The hierarchy claims that Level 3 is not merely “more of Level 2”: it can change how deep renormalization goes, not only how large the step is. That pressure yields Proposal 2: a separate surplus channel with its own Feigenbaum depth R_3 , and an optional jump term.

Proposal 1 is the candidate for deployment: conservative, ablatable, and computationally cheap. Proposal 2 is the candidate for structural comparison with equation (2), testing whether a dual-channel geometry aligns better with nested f_3 on controlled synthetic systems. If Proposal 1 fails to help under real hyper-persistence, there is no operational case for the more aggressive Proposal 2. If Proposal 1 helps but still under-represents Level-3 geometry, Proposal 2 is the natural next experiment rather than a first field change. Sections 3–4 formalize both constructions; Section 5 freezes the shared activation logic; Section 8 states the operational priority.

2. Shared base notation

The legacy base increment (without Level 3) is defined by

$$\Delta t_k^{\text{base}} = \begin{cases} \delta^{-k_{\text{run}}} \cdot |\tau_s(k)| \cdot \Delta t_0 & \text{if } |\tau_s(k)| < \tau_{\text{ch}}, \\ \Delta t_{\text{out}} & \text{otherwise (unit or null advance, pipeline-dependent),} \end{cases} \quad (4)$$

where k_{run} is the depth of the current chaotic run (or a bounded local renormalization counter). The gate g may be piecewise or sigmoidal; the generic form $g(\tau_s) \in [-1, 1]$ is retained.

Normalization of excess³ (for numerical stability across datasets):

$$\tilde{e}_3(k) = \frac{\text{excess}^3(k) - \mu_{e,B}}{\sigma_{e,B} + \varepsilon}, \quad (5)$$

with $(\mu_{e,B}, \sigma_{e,B})$ estimated exclusively on a baseline window B fixed a priori (or on a calibration set that does not overlap the evaluation event). $\varepsilon > 0$ is a small stabilizer. A rank-based alternative (more robust under heavy tails) is

$$\tilde{e}_3^{\text{rk}}(k) = 2 \cdot \hat{F}_B(\text{excess}^3(k)) - 1 \in [-1, 1], \quad (5')$$

where $\hat{F}_B$ denotes the empirical cumulative distribution function on B .

Ordinal persistence (as in the RQA / hyper-persistence literature):

$$P(k) = \text{length of the current run with } |\tau_s| < \tau_{\text{ch}}. \quad (6)$$

Nested Level-3 mass (diagnostic; not the sole integration path):

$$f_3(k) = \frac{\alpha_3(\lambda) \Phi_3^*(k)}{\sum_{\ell=1}^3 \alpha_{\ell}(\lambda) \Phi_{\ell}^*(k) + \varepsilon}. \quad (7)$$

3. Proposal 1 (conservative): multiplicative surplus gain

Under Proposal 1, the base advance remains determined by τ_s , the gate g , and the state factor, as in the legacy pipeline. Φ_3 enters only as a secondary scale factor on that already-decided increment: when irreducible surplus is absent or stagnant, $\Gamma_3 = 1$ and the clock is unchanged; when surplus is present and changing, $\Gamma_3 > 1$ and the same base step is stretched. Setting $\beta = 0$ is therefore an exact ablation: it removes the surplus factor without altering the primary clock.

3.1 Formulation

Equations (1) and (4) are left structurally unchanged. Φ_3 enters solely as a gain factor $\Gamma_3 \geq 1$ (or ≥ 0 if a braking regime is admitted) applied to the increment already filtered by g and by the state factor:

$$t_{k+1} = t_k + \Delta t_k^{\text{base}} \cdot g(\tau_s(k)) \cdot \alpha(\text{state}_k) \cdot \Gamma_3(k) \quad (\text{P1})$$

3.2 Definition of Γ_3

Let $A_3(k) \in \{0, 1\}$ denote the Level-3 activation gate (Section 5). Define

$$\Gamma_3(k) = 1 + \beta \cdot A_3(k) \cdot \psi(\tilde{e}_3(k)), \quad (8)$$

where $\beta \geq 0$ is a sensitivity hyperparameter (default range $\beta \in [0.25, 1.0]$, fixed a priori) and ψ is a monotone bounded map, for example

$$\psi(x) = \text{softplus}(x) = \log(1 + e^x), \quad \text{or} \quad \psi(x) = \max(0, x). \quad (9)$$

Argument of ψ :

Mode	Argument of ψ	Intended regime
level (default)	$ \tilde{e}_3 $ if absolute form; signed softplus otherwise	Elevated surplus magnitude
change	$ \Delta \tilde{e}_3 $ (or signed first difference)	Reorganization with a decrease in surplus (G1)
both	mean of level and change arguments	Compromise under mixed dynamics

The absolute-delta form adopted as default (consistent with G1, in which surplus may decrease) is

$$\Gamma_3^{\text{abs}}(k) = 1 + \beta \cdot A_3(k) \cdot \psi(|\Delta \tilde{e}_3(k)|) \quad \text{or} \quad \psi(|\tilde{e}_3(k) - \tilde{e}_3^{\text{ref}}|), \quad (8')$$

where $\tilde{e}_3^{\text{ref}}$ is the baseline median. Absolute-delta privileges change in surplus over chronic level: in hyper-persistent stretches, excess³ can remain elevated without marking a transition, whereas reorganization—including some G1 paths—often appears as a movement of surplus (sometimes a drop). The implementation exposes the absolute-delta switch and `psi_mode` $\in \{\text{level}, \text{change}, \text{both}\}$ so that sensitivity to this choice remains transparent.

3.3 Modulation of the Φ_3 weight

Quantity	Role
excess ³ (k)	Continuous proxy for Φ_3 (equation 3)
$\tilde{e}_3(k)$ or $ \Delta \tilde{e}_3 $	Dimensionless scale
$A_3(k)$	Suppresses trivial activation under hyper-persistence
β	Global channel intensity (single pre-specified scalar)
$\alpha_3(\lambda)$ (optional)	May enter A_3 or a compound weight $\beta_{\text{eff}} = \beta \cdot \alpha_3(\lambda) / \alpha_{3,0}$

Compatibility with equation (2). Proposal 1 does not replace nested RECD; it defines a map F on the legacy pipeline. Optionally, one may report in parallel

$$\Delta\text{RECD}^{\text{nested}}(k) = \sum_{\ell=1}^3 \alpha_{\ell}(\lambda) \Phi_{\ell}^*(k)$$

and require positive correlation between $\Gamma_3(k) - 1$ and $f_3(k)$ in synthetic chaotic regimes.

Marginal contribution of Φ_3 to the clock (generalization of ρ_3 from Proposal 2 to Proposal 1):

$$\rho_3^{\text{marg}}(k) = \frac{|\Delta t_k^{(P1)} - \Delta t_k^{\text{base}}|}{|\Delta t_k^{(P1)}| + \varepsilon} = \frac{|\Gamma_3(k) - 1| \cdot |\Delta t_k^{\text{base}} g \alpha|}{|\Delta t_k^{(P1)}| + \varepsilon}. \quad (8'')$$

This quantity supports ablation and direct comparison with ρ_3 under Proposal 2.

3.4 Predicted dynamical consequences

1. Selective acceleration. In regime C with $A_3 = 1$ and elevated surplus (or large $|\Delta e_3|$), t_k advances more rapidly than base RECD at equal τ_s .
2. Absence of discontinuous jumps when ψ and $\tilde{e}_3$ are smooth: the clock remains piecewise absolutely continuous (only the local slope changes).
3. Improved pre-critical discrimination. In cardiac series, if Δexcess^3 precedes τ_s , Γ_3 may advance critical-mass accumulation and improve lead time on T_{RECD} .
4. Exact ablation. Setting $\beta = 0$ or $A_3 \equiv 0$ recovers the legacy RECD exactly.

3.5 Implementation (pseudocode)

for k in time:

```

tau = systemic_tau(window_k)
e3  = excess3(window_k)          # fixed 0.6/0.4
e3n = normalize_vs_baseline(e3)
dt  = base_interval(tau, run_depth)
g   = gate(tau)
a   = state_factor(state_k)      # C/D/T
A3  = activation_level3(...)      # Section 5
Gamma = 1 + beta * A3 * softplus(e3n) # or abs-delta form
t[k+1] = t[k] + dt * g * a * Gamma

```

Computational cost is of the same order as excess^3 evaluation per window and is already incurred whenever excess^3 is computed for early warning.

4. Proposal 2 (experimental): dual channel with depth-dependent compression

Proposal 1 only scales a step already fixed by τ_s . Proposal 2 allows Φ_3 to contribute an additive component of the step and to change the renormalization depth of that component. Channel C_{12} is the pair-relational advance of the standard construction; channel C_3 is discrete time attributable to irreducible surplus, compressed more aggressively the longer surplus remains active. That geometry is closer in spirit to nested equation (2), but it is also easier to overfit and harder to reverse-engineer in operational code. Hence the experimental status: evaluate on synthetic systems first; do not deploy until Proposal 1 has shown domain benefit.

4.1 Formulation

Clock advance is decomposed into two additive channels:

- Pair-relational channel C_{12} : governed by τ_s , P , Φ_1 , and Φ_2 (as in the standard construction).
- Surplus channel C_3 : governed by excess^3 and active only under the strict criteria of Section 5.

When channel 3 is active, the effective renormalization depth increases (stronger Feigenbaum compression), and optional bounded structural jumps may be admitted.

$$\begin{aligned} \Delta t_k^{(12)} &= \delta^{-k_{12}} \cdot h_{12}(\tau_s(k), \Phi_1(k), \Phi_2(k)) \cdot g(\tau_s(k)) \cdot \alpha(\text{state}_k), \\ \Delta t_k^{(3)} &= A_3(k) \cdot \delta^{-(k_{12} + \kappa \cdot R_3(k))} \cdot h_3(\tilde{e}_3(k), f_3(k)), \\ t_{k+1} &= t_k + \Delta t_k^{(12)} + \Delta t_k^{(3)} + A_3(k) \cdot J \cdot 1\{\text{jump condition}\}. \end{aligned} \quad (\text{P2})$$

4.2 Components

Surplus depth (bounded accumulator, not a free exponent):

$$R_3(k+1) = \min(R_{\max}, (R_3(k) + 1) \cdot A_3(k)), \quad R_3 \leftarrow 0 \text{ if } A_3 = 0. \quad (10)$$

Thus sustained runs of active Level 3 deepen the compression $\delta^{-(\cdot)}$ analogously to k_{run} on channel 12, but only when surplus is non-trivial.

Amplitude of channel 3:

$$h_3(\tilde{e}_3, f_3) = \eta \cdot \psi(|\tilde{e}_3|) \cdot (1 + \gamma_f f_3), \quad (11)$$

with $\eta > 0$, $\gamma_f \geq 0$ pre-specified; default $\gamma_f = 0$ if f_3 is unused.

Structural jump (optional; initially disabled):

$$\text{jump condition} \iff A_3(k) = 1 \wedge \tilde{e}_3(k) > \theta_{\text{jump}} \wedge \Delta \tilde{e}_3(k) > \theta_{\Delta} \wedge P(k) \geq P_{\min}. \quad (12)$$

$J \geq 0$ is a fixed jump size or $J = j_0 \cdot \psi(\tilde{e}_3)$. Under a smoothness constraint, $J = 0$.

4.3 Modulation of the Φ_3 weight

Under Proposal 2 the weight of Φ_3 is not a single scalar β ; it enters through

1. the gate A_3 (binary or soft in $[0, 1]$);
2. the magnitude h_3 ;
3. the depth R_3 (nonlinear effect via δ^{-R_3});
4. optionally the nested mass f_3 .

Level 3 may therefore dominate the mathematical regime of the clock (irregularity, scale jumps), not only its local slope.

4.4 Predicted dynamical consequences

1. Temporal bimodality. Intervals governed solely by C_{12} (pair-dominated chaos) versus intervals governed by $C_{12} + C_3$ (chaos with surplus), with distinct speed and roughness of T_{RECD} .
2. Additional hierarchical compression on runs of active Φ_3 : shorter intervals as R_3 grows (up to $R_{\max}$).
3. Jumps (if $J > 0$): discrete clock events aligned with surplus transitions—candidates for Level-3 ticks in early-warning applications.
4. Increased robustness to unstructured hyper-persistence. If $A_3 = 0$ during a long hyper-persistent core with stagnant excess³, the clock does not accumulate spurious Φ_3 contribution; if excess³ reorganizes within the core, C_3 activates (a natural complement to RQA).

4.5 Relation to the nested equation (2)

Proposal 2 may be interpreted as a nonlinear realization of equation (2):

- C_{12} corresponds to the terms $\alpha_1 \Phi_1 + \alpha_2 \Phi_2$ after gating and Feigenbaum compression;
- C_3 corresponds to the term $\alpha_3 \Phi_3^*$ with an independent renormalization schedule.

Consistency prediction. On coupled logistic maps (pre-chaos versus chaos), the fraction

$$\rho_3 = \frac{\sum_k |\Delta t_k^{(3)}|}{\sum_k (|\Delta t_k^{(12)}| + |\Delta t_k^{(3)}|)}$$

is expected to rise in chaos analogously to f_3 in the 2026 framework (on the order of +0.25 absolute under the reference protocol, subject to re-estimation).

5. Hardened Level-3 activation criteria

Objective. Φ_3 influences the clock only when non-trivial surplus is present and the regime context is compatible.

In tropical dengue series and in many Holter recordings, the system spends a large fraction of time in band C (hyper-persistence). If A_3 fired whenever $|\tau_s| < \tau_{ch}$, then $\Gamma_3 > 1$ almost continuously: the gain would stop carrying information about transitions and would become a permanent rescaling of the clock. Synthetic checks confirm the risk: a permissive gate can activate on more than half of a null series (Section 6.3). Hardened activation is therefore required to keep Φ_3 from collapsing into a synonym of band C.

The criteria below implement four requirements: (A) chaotic or approach context; (B) surplus that is high and moving; (C) surplus not reducible to pairwise coincidence; (D) confirmation at two timescales so that a single noisy spike cannot open the channel.

5.1 Primary gate $A_3(k)$

Set $A_3(k) = 1$ if and only if all of the following conditions hold (strict version; the soft version combines scores in $[0, 1]$):

(A) Chaotic or approach regime

$$|\tau_s(k)| < \tau_{ch} \quad \text{or} \quad P(k) \geq P_{arm} \quad \text{or} \quad \text{state}_k \in \{C, T\}. \quad (\text{A})$$

P_{arm} (default 3) is smaller than the hyper-persistent core threshold ($P \geq 7$).

(B) Surplus level and (B') change (both required by default)

$$|\tilde{e}_3(k)| > \theta_e^{\text{eff}} \quad \text{and} \quad |\Delta \tilde{e}_3(k)| > \theta_\Delta. \quad (\text{B} + \text{B}')$$

The change condition is essential: in synthetic G1, surplus may decrease under a shared latent factor; a level-only gate would fail or activate in the wrong regime. Defaults: $\theta_e = 1.25$ (z-score units), $\theta_\Delta = 0.40$; change is required.

Adaptive threshold (noisy series):

$$\theta_e^{\text{eff}} = \theta_e \cdot (1 + \kappa \cdot \min(\text{CV}_B, 2)), \quad (\text{B}_{\text{adapt}})$$

with $\text{CV}_B = \sigma_B / (|\mu_B| + \epsilon)$ estimated only on baseline and $\kappa = 0.5$ by default. This construction reduces false positives when baseline excess³ is highly variable.

(C) Residual synergy fraction

Rather than the bare ratio $\text{excess}^3 / (\Phi_1 + \Phi_2)$, an approximate residual fraction of Level-3 mass is used:

$$r_{\text{syn}}(k) = \frac{s_B(k)}{s_B(k) + \Phi_1(k) + \Phi_2(k) + \varepsilon} \in (0, 1), \quad r_{\text{syn}}(k) > \theta_{\text{res}}, \quad (\text{C})$$

where $s_B = |\tilde{e}_3|$ (or signed softplus). Default $\theta_{\text{res}} = 0.20$. If Φ_1 and Φ_2 are unavailable, the proxy $\Phi_1 + \Phi_2 \equiv 1$ may be substituted (same functional form; recalibratable). Optionally, $f_3 > \theta_f$ may also be required.

(D) Dual scale: short detection and medium persistence

$$\begin{aligned} \#\{j \in [k - L_s + 1, k] : (\text{B}) \text{ and } (\text{C})\} &\geq L_{s,\min}, \\ \#\{j \in [k - L_m + 1, k] : (\text{A}), (\text{B}), (\text{C})\} &\geq L_{m,\min}. \end{aligned} \quad (\text{D})$$

Defaults: $L_s = 3$, $L_{s,\min} = 2$ (detection); $L_m = 7$, $L_{m,\min} = 4$ (surplus persistence). Neither a single spike nor a brief noisy excursion is sufficient.

(E) Optional — RQA consensus under hyper-persistence

On datasets with a hyper-persistent core (SJU-3, DengAI San Juan):

$$A_3^{\text{RQA}}(k) = A_3(k) \cdot 1\{\text{LAM}_{\text{core}}(k) > \lambda_{\text{LAM}} \vee \Delta\text{TT}(k) > \theta_{\text{TT}}\}. \quad (\text{E})$$

5.2 Soft gate (operational default for Proposal 1)

$$A_3^{\text{soft}}(k) = \sigma(c_A + c_B + c_C + c_D - s_0) \in (0, 1), \quad (13)$$

where c_B is the product of level and change scores, c_C is a logistic transform of r_{syn} , c_D averages the two temporal scales, and $s_0 = 3.25$ (logistic offset). Soft values below 0.25 are set to zero to suppress residual activation under the null.

5.3 Exclusion rules

Φ_3 must not activate under any of the following conditions:

- occupancy of the entire C regime solely because $|\tau_s| < 0.41$;
- a single excess³ spike without dual-scale confirmation (D);
- elevated excess³ level without change (low $|\Delta\tilde{e}_3|$)—characteristic of “flat” hyper-persistence;
- pure pairwise coincidence: low r_{syn} even when $|\tau_s|$ lies in the chaos band;
- windows with $d = 2$: channel 3 is disabled and the disability is reported;
- threshold calibration on the same event used for evaluation (data leakage).

6. Validation design and results

6.1 Principles

1. Ablation first. Proposal 1 or Proposal 2 is compared with base RECD ($\beta = 0$ / $A_3 \equiv 0$).

2. Phase-shuffle nulls are applied to the primary statistic $|\Delta T|$ or lead time, not solely to isolated excess³.
3. Pre-registration of hyperparameters ($\beta, \theta_e, L_{\min}, \dots$) precedes inspection of outbreak or ventricular-fibrillation labels.
4. Detection improvement (lead, precision, false-alarm rate) is reported separately from change in clock geometry (fractal dimension of T , roughness, ρ_3).

6.2 Synthetic validation design

Synthetic arms exist to answer questions that real data cannot isolate cleanly. S0 asks: does the gate stay quiet when there is no true surplus structure? S1 asks: when reorganization is known, does the gain track nested mass even if surplus falls? S2 asks: does chaos elevate Φ_3 contribution relative to pre-chaos? S3–S4 tighten specificity against pairwise-only coupling and flat hyper-persistence.

Arm	Design	Primary prediction	Human question
S0	Independent channels / G0-like	Hard $A_3 \approx 0$; low soft rate; non-spurious ΔT_{P1}	Does the null stay null?
S1	Latent factor + reorganization (G1; surplus may decrease)	$ \Delta e_3 $ drives A_3 ; high $\text{corr}(\Gamma_3 - 1, f_3)$	Does change-based activation catch true reorg?
S2	Pre-chaos versus chaos	$\mathbb{E}[\Gamma_3]$, ρ_3 , and A_3 elevated in chaos	Is Level 3 selective for chaos?
S3	Pairwise coupling only (G3)	Intermediate activation; must not exceed S1	Pairs alone must not look like Φ_3
S4	Flat hyper-persistence versus reorganization	Discrimination via (B') and (D); (E) reduces false alarms	Flat chaos $\neq$ reorganizing chaos

Metrics. A_3 rate (soft > 0.5 and hard); T_{final} under base / P1 / P2; mean Γ_3 ; ρ_3 (P2) and ρ_3^{marg} (P1); $\text{corr}(\Gamma_3 - 1, f_3)$; noise robustness at 5–15%. Soft A_3 is the operational continuous score used inside Γ_3 ; hard A_3 is the binary gate used for lead/FAR tables—two views of the same criteria, not two competing theories.

6.3 Synthetic results

Synthetic runs employ the shared activation defaults listed in Appendix A.

Arm	A_3 soft >0.5	A_3 hard	mean Γ_3	ρ_3 P2	ρ_3^{marg} P1	$\text{corr}(\Gamma - 1, f_3)$	ΔT_{P1}
S0	0.003	0.000	1.015	0.016	0.01	0.02	+4.2
S1	0.230	0.200	1.893	0.201	0.18	0.988	+1.9
S2 prechaos	0.000	0.000	1.026	0.064	0.02	−0.01	+1.3
S2 chaos	0.195	0.165	1.421	0.322	0.21	0.821	~0†

†In chaos, Δt^{base} is already Feigenbaum-compressed: absolute ΔT is small; ρ_3 and Γ_3 capture the Φ_3 contribution.

Full numerical tables and T_{final} series use seed base = 10.

Interpretation.

1. S0 (null). Hard $A_3 = 0$, soft rate $\approx 0.3\%$, mean $\Gamma_3 \approx 1.02$, $\Delta T_{P1} \approx +4$ ($\sim 1\%$ of the base clock).

The hardened gate respects the null. A more permissive configuration (without surplus-change or residual-synergy requirements) raised mean A_3 to ~ 0.55 on the same series, which motivates the present hardening.

2. S1 (latent / change). Selective activation ($\sim 20\text{--}23\%$) and near-unit correlation between $\Gamma_3 - 1$ and f_3 . The Φ_3 channel aligns with nested mass under true reorganization, including a decrease in surplus.
3. S2. The chaotic regime elevates Γ_3 (1.42 versus 1.03) and ρ_3 (0.32 versus 0.06). Pre-chaos does not activate A_3 .
4. Noise (5–15%). Gradual degradation without collapse of the ordering $S0 \ll S1 / S2\text{-chaos}$ (Figure, noise-robustness panel).

Pre-specified acceptance criteria.

Criterion	Observed result
S0 hard $A_3 \lesssim 8\%$	Met (0.000)
S0 mean Γ_3 near 1	Met (1.015)
S2 chaos $E[\Gamma_3] >$ pre-chaos	Met (1.421 > 1.026)

On a contrast series with a surplus burst superimposed on a chaotic core, the rate $A_3 > 0.5$ reaches ≈ 0.95 inside the burst and falls to ≈ 0.25 on the burst-free core (versus ≈ 0.72 under a globally permissive gate). The hardened gate therefore discriminates local reorganization from flat chaos.

6.4 Paired empirical analyses (Holter and DengAI): limits of the legacy clock

The empirical design is paired: the same Proposal 1 core and β run on Holter recordings from the PhysioNet Sudden Cardiac Death Holter Database (SDDDB) and Normal Sinus Rhythm Database (NSRDB) [4, 14] (seconds-to-hours, RR, pre-VF) and on DengAI San Juan/Iquitos weekly climate-incidence series [15] (outbreak peaks). The systems are not physically similar; the comparison tests whether operational failure modes of the legacy clock are domain-specific or generic.

These limitations are largely generic across domains. Both the Holter and DengAI series spend most of the evaluation window in band C. Under equation (4), long chaotic runs deepen Feigenbaum compression $\delta^{-k_{\text{run}}}$, so Δt^{base} collapses toward zero. Multiplying a near-zero base by $\Gamma_3 > 1$ still yields a near-zero step: the gain cannot inflate a clock that the legacy construction has already frozen. Many Holter and DengAI rows therefore show $\Delta T \approx 0$ even when A_3 or excess³ move. The exception (Iquitos 2004) is instructive: chaos occupancy falls to $\approx 88\%$, active Δt^{base} reappears, and legacy $\Delta T = +1.24$.

The paired design therefore evaluates two operational questions: whether hardened A_3 remains selective under real hyper-persistence (alarm behaviour), and under what conditions Γ_3 can displace the legacy index T (clock behaviour).

Proposal 1 uses the same activation core and β in both domains. Sample sizes are small relative to a full cohort study (Section 6.5). Thresholds for A_3 were fixed a priori from synthetic S0 and one-at-a-time sensitivity analyses, and were not re-optimized on ventricular-fibrillation or outbreak labels.

6.4.1 Contrastive protocol

	Holter (cardiology)	DengAI (epidemiology)
Data	SDDDB 30/31/35 + NSRDB 16265/16272	San Juan and Iquitos training series; top-3 peaks $\geq$ P80, separation $\geq$ 26 weeks
Proxy / W / $d_{\min}$	$[z(\text{RR}), z(\Delta\text{RR})]$, $W=101$, stride 5, $\theta_3=0.08$, $d_{\min}=2$	$[\text{cases}, T, \text{precip}, \text{RH}]$, $W=13$, stride 1, $\theta_3=0.10$, $d_{\min}=3$
Events / controls	pre-VF versus NSRDB	40 weeks pre-peak (basal 13 + approach 12) versus inter-epidemic maxima $<$ P60
P1 / ablation	Hardened activation; $\beta \in \{0, 0.5\}$	identical

Figures. Holter pre-event series, baseline→approach rates and lead, and false-alarm rate; DengAI analogues with weekly time axis and outbreak peaks marked (Appendix D).

6.4.2 Results by domain

Holter — events and controls

Rec	type	chaos%	A_3 hard bas→app	Γ_3 app	ΔT legacy	ΔT Holter [†]	Lead A_3 / e^3
30	event	100%	0.000 → 0.000	1.012	≈ 0	+89	— / 3.1 h
31	event	100%	0.001 → 0.000	1.014	≈ 0	+197	— / 6.9 h
35	event	100%	0.004 → 0.003	1.054	≈ 0	+625	5.0 h / 8.0 h
16265	control	100%	rate 0.014	1.118	≈ 0	+1402	FAR $\approx 21.6/24\text{h}$
16272	control	100%	rate 0.019	1.134	≈ 0	+1491	FAR $\approx 25.6/24\text{h}$

[†] ΔT Holter denotes a diagnostic clock $\Delta t = \Delta t_0 g(\tau) \Gamma_3$ without Feigenbaum depth; it is not the default of equation (1). It is reported only as a counterfactual: the displacement Γ_3 would produce if the base interval were not compressed to zero. The operational claim about legacy ΔT uses the full equation (4) construction.

Holter aggregates. Hard A_3 sensitivity 1/3 versus absolute-z excess³ 3/3; active $\Delta t^{\text{base}} \sim 0.1\%$.

DengAI — outbreaks and controls

ID	type	Peak	chaos%	A_3 hard bas→app	Γ_3 app	ΔT legacy	Lead A_3 / e^3
SJ-O1-1994	outbreak	461	100%	0.00 → 0.00	1.11	≈ 0	— / —
SJ-O2-1998	outbreak	329	98%	0.00 → 0.00	1.03	≈ 0	— / —
SJ-O3-2007	outbreak	170	100%	0.07 → 0.23	1.40	≈ 0	— / 4 w
IQ-O1-2004	outbreak	116	88%	0.00 → 0.15	1.39	+1.24	11 w / 23 w
IQ-O2-2008	outbreak	58	100%	0.00 → 0.62	2.38	≈ 0	25 w / 26 w
IQ-O3-2008	outbreak	63	100%	0.00 → 0.69	2.38	≈ 0	10 w / 15 w
controls (4)	inter-ep.	—	95–100%	rate 0–0.24	1.12–1.56	≈ 0	mean FAR 1.62/52w

DengAI aggregates. Hard A_3 sensitivity 3/6 (all Iquitos) versus excess³ 4/6; mean legacy ΔT +0.207 (driven by IQ-2004); active $\Delta t^{\text{base}} \sim 7.3\%$; mean outbreak chaos 97.6%.

6.4.3 Cross-domain synthesis

	Holter	DengAI	Cross-domain interpretation
Band C $\rightarrow$ legacy ΔT	chaos 100%; $\Delta t^{\text{base}} \sim 0.1\%$; $\Delta T \approx 0$	chaos $\sim 98\%$; $\Delta t^{\text{base}} \sim 7\%$; $\Delta T \approx 0$ except IQ-2004 (+1.24)	C saturation is not Holter-specific. Γ_3 displaces T only if $\Delta t^{\text{base}} \neq 0$.
Gate A_3 / lead / FAR	hard $\lesssim 0.02$; sens. 1/3 vs e^3 3/3; FAR $\sim 24/24\text{h}$	bas $\rightarrow$ app in Iquitos; sens. 3/6 vs e^3 4/6; FAR $\sim 1.6/52\text{w}$	(B) $\cap$ (D) restrict activation; A_3 is stricter than excess ³ ; event/outbreak specificity remains to be demonstrated.
Ablation $\beta=0$	$\max \Delta T = 0$	$\max \Delta T = 0$	The Φ_3 channel is ablatable in both domains.

Cross-domain conclusions.

1. Proposal 1 is implementable and auditable in two empirical domains with a shared computational core and parameters fixed a priori.
2. Hardened A_3 functions as a selectivity gate. Hyper-persistence of regime C does not imply hyper-persistence of Φ_3 alarms. On Holter recordings, hard activation remains scarce. In Iquitos, activation discriminates baseline from approach when surplus and dual-scale persistence are jointly present.
3. The legacy clock is not a universal vehicle for lead time. Under sustained occupancy of band C, Feigenbaum compression drives Δt^{base} toward zero; multiplication by Γ_3 then leaves T unchanged. IQ-2004 illustrates the conditional mechanism: chaos 88%, active Δt^{base} about 24%, $\Delta T = +1.24$, A_3 lead 11 weeks. On Holter data, A_3 primarily supplies a conditioned alarm rather than inflation of T . The clock without Feigenbaum depth is reported only as an auxiliary diagnostic index.
4. Neither reduction in false-alarm rate nor operational superiority is established. Under the fixed thresholds used, controls activated as often as—or more often than—events or outbreaks. Extreme case-count peaks (San Juan 1994 and 1998) do not activate hard A_3 : multivariate ordinal surplus is not reducible to univariate incidence magnitude.

Further empirical extensions include a domain-specific clock or redefinition of condition (A); DengAI proxy-sensitivity analyses; an SDDB cohort with $N \geq 10$ and surrogates; SJU-3; and RQA condition (E) once the domain clock is stabilized.

6.5 Extended empirical validation

Dataset	Validation objective
DengAI (extension)	Additional peaks; surrogates; covariate-proxy sensitivity; outbreak threshold
SJU-3 (12 traps)	Effect of Φ_3 on the clock under core hyper-persistence ($\sim 69.6\%$ occupancy)
SDDB Holter ($N \approx 10$ high-quality)	Cohort lead and FAR; domain clock versus legacy; surrogates
Negative control	Full NSRDB; temporal label shuffle; long DengAI baseline periods

Comparison protocol (per event).

1. Compute T^{base} , $T^{(P1)}$, and (optional diagnostic) a clock without Feigenbaum depth.
2. Define alarms on hard and soft A_3 and, in parallel, on the absolute z-score of excess³.
3. Report lead, false-alarm rate, and phase-shuffle p -values.

4. Ablation: $\beta = 0$; removal of (B), (C), or (D) individually.

6.6 Pre-specified success criteria

Operational success of Proposal 1 is declared if, in at least two of three domains (synthetic S2, dengue, cardiology),

1. the median lead of $T^{(P1)}$ is at least that of T^{base} with false-alarm non-inferiority ($+\leq 5$ absolute percentage points), or
2. the false-alarm rate is significantly lower at comparable lead, and
3. on S0, the A_3 activation rate is bounded and $T^{(P1)}$ exhibits no spurious superiority.

Results relative to these criteria. Criterion (3) is met on synthetic S0. Criteria (1)–(2) are not satisfied by the paired empirical analyses of Section 6.4. Those analyses do establish: (i) selectivity of A_3 under real hyper-persistence; (ii) dependence of legacy ΔT on escape from band C; (iii) exact $\beta = 0$ ablation; (iv) event- and outbreak-level specificity under fixed thresholds remains to be demonstrated.

Structural success of Proposal 2 further requires that ρ_3 separate S2 regimes consistently with nested f_3 . Proposal 2 is assessed on synthetic S2 only.

7. Risks and limitations

Risk	Description	Mitigation
Double counting	excess ³ correlated with τ_s ; Φ_3 may add no new clock information	Condition (C); ablation; report $I(\Gamma_3; \tau_s)$
Hyper-persistence	Near-continuous activation under chronic band C	(D)+(E); A_3 on Δe_3 ; (B) $\cap$ (D) (Section 6.4)
Feigenbaum / band C	Chronic $ \tau < \tau_{\text{ch}}$ freezes $\Delta t^{\text{base}} \approx 0$; legacy ΔT uninformative	Domain clock without depth; redefine (A); non-saturating proxy; lead via alarms
Threshold overfitting	Multiplicity of θ	Fixed defaults; one-at-a-time sensitivity; no fit on test events
Degenerate $d = 2$	Order-3 surplus ill-defined	Disable channel 3 if $d < 3$
Surrogate cost	Online p -values for (B) are expensive	Online z -score / baseline CDF; offline surrogates
Sign of Δe_3	G1 can yield $\Delta e_3 < 0$ under true reorganization	Prefer absolute-delta (8') for early warning
R_3 instability (P2)	Over-aggressive exponential compression	Low R_{max} (2–4); $J = 0$ initially
Partial-information confusion	excess ³ misread as a complete causal atom	Pre-specified hybrid proxy, not complete PID
Two formalisms	Legacy (1) versus nested (2)	Report both; P1 as minimal bridge
Jump parameters	$J > 0$ may appear ad hoc	Keep $J = 0$ until S4 and extended criteria are met

Conceptual limitation. excess³ is not a complete partial-information decomposition. Its incorporation into the clock improves the dynamics of the index; it does not convert the proxy into an axiomatic theory of synergistic information. Causal interpretation of the data-generating process remains deliberately cautious.

8. Operational priority

8.1 Operational default (Proposal 1)

Proposal 1 (conservative) is adopted as the default upgrade of the operational pipeline, with:

- the absolute-delta form (8') as the default gain;
- the soft gate (13) with conditions (A)–(D);
- optional extension (E) on hyper-persistent datasets;
- pre-specified β (for example, 0.5) and sensitivity sweep $\beta \in \{0, 0.25, 0.5, 1.0\}$.

The default is motivated by five properties:

1. Dynamical closure with minimal structural change. Φ_3 is no longer confined to a parallel diagnostic: it modifies Δt_k explicitly and in an ablatable manner.
2. Falsifiability. Setting $\beta = 0$ recovers the standard operational construction without the Φ_3 channel.
3. Controlled behaviour under hyper-persistence. When A_3 tracks surplus change rather than mere occupancy of band C, trivial activation is limited.
4. Computational feasibility. The construction is implementable on existing nested-RECD, dengue, and cardiology pipelines at $O(1)$ additional cost after excess³ has been computed.
5. Compatibility. The proposal is consistent with nested RECD (2) and with six-mode RQA (Φ_3 -gain may enter as an additional mode or clock weight, not as a substitute for LAM/TT).

Under chronic band C, Feigenbaum compression can freeze Δt^{base} ; Γ_3 then cannot displace T until the base reopens. In that regime, A_3 should be evaluated as a conditioned alarm, and a domain clock without depth may be considered (Section 6.4, Section 7). The excess³ weights 0.6/0.4 remain fixed so that clock effects are not confounded with retuning of the proxy on the same evaluation events.

8.2 Status of Proposal 2

Proposal 2 is an experimental dual-channel extension and is not the operational default. Dual-channel depth and optional jumps change the shape of discrete time, not only its scale. That construction is useful for testing consistency with nested f_3 , but it multiplies free parameters and makes field failures harder to localize. Proposal 2 is therefore considered only after Proposal 1 shows domain benefit and nested mass f_3 still diverges from the operational clock geometry (Section 4; synthetic S2).

- Structure. Dual channel $C_{12} + C_3$, depth R_3 , and metric ρ_3 aligned with nested f_3 (evaluated on synthetic suite S2).
- Scope of evaluation. Synthetic systems only; $J = 0$ and $R_{\text{max}} = 3$.
- Condition for further development. Jumps and deeper compression are considered only after Proposal 1 demonstrates domain benefit under the criteria of Section 6.6.
- Surveillance applications. Proposal 2 is not applied to epidemiological or cardiac deployment until those criteria are satisfied.

8.3 Relative priority among related developments

Development	Relative priority
Integration of Φ_3 into RECD dynamics (P1)	Primary operational priority
Adaptive RQA and cross-recurrence communities	Parallel; couples via condition (E)
Additional out-of-sample validation of excess ³	Informs thresholds of (B)
Complete ordinal partial-information decomposition	Longer-term; independent of P1 deployment

8.4 Open empirical steps

Priority empirical extensions include: an SDDDB cohort analysis with a domain-specific clock; DengAI extension with surrogate ensembles [11]; SJU-3; and RQA condition (E) [3, 13] once the domain clock is stabilized in cardiology and dengue. Shared numerical defaults are listed in Appendix A.

9. Conclusion

Prior work on Systemic Tau and the Discrete Extramental Clock has established ordinal relational metrics and a base discrete-time construction [1, 5], excess³ as a continuous Level-3 proxy [2], Holter evidence of pre-VF ordinal reorganization [4], and recurrence-based refinement under hyper-persistence [3, 13]. The unresolved operational question is how Φ_3 should enter the construction of the index t_k in equation (1), while remaining consistent with the nested mass representation (2).

We develop two complementary answers because no single map is forced by the theory. Proposal 1 keeps the legacy clock and multiplies its increment by a surplus-gated gain—the smallest change that makes Φ_3 dynamical and ablatable. Proposal 2 gives Φ_3 its own channel and depth—the construction needed to test structural alignment with nested mass, but not yet justified for field use. Shared hardened activation criteria prevent band-C hyper-persistence from turning either construction into a permanent rescaling.

The principal results are:

1. a structural diagnosis of the gap between parallel use of Φ_3 and a generative contribution to the operational clock;
2. a multiplicative, ablatable Proposal 1 suitable for operational pipelines;
3. a dual-channel Proposal 2 with depth-dependent compression (experimental; synthetic evaluation only);
4. hardened activation criteria based on surplus change, residual synergy, dual-scale persistence, and adaptive θ_e ;
5. synthetic results (S0–S2) with null A_3 on S0, discrimination of chaos from pre-chaos, and high $\Gamma_3 \leftrightarrow f_3$ alignment on S1;

6. paired Holter and DengAI analyses: A_3 limits alarm hyper-persistence; Feigenbaum compression and band-C occupancy restrict legacy ΔT in both domains except for partial escape (IQ-2004); event- and outbreak-level specificity under fixed thresholds remains to be demonstrated.

In summary, Proposal 1 is auditable and implementable across domains; the hardened A_3 gate successfully controls hyper-persistence; the legacy clock advances only when there is partial escape from the chaos band; and domain-specific validation remains necessary before any operational claims can be made. Displacement of T still requires escape from band C (or a domain clock that does not freeze Δt^{base}). Cohort-scale validation remains a prerequisite for deployment.

excess³ is treated throughout as a pre-specified hybrid proxy rather than as a complete partial-information decomposition. Φ_3 enters the clock solely through falsifiable, ablatability maps F .

Appendix A — Default parameters

Symbol / field	Default	Notes
Synthetic W	13	S0–S2 generators
DengAI W	13	stride 1; $\theta_3 = 0.10$
Holter W	101	stride 5; $\theta_3 = 0.08$
(CCTP)		
m	3	delay 1
(Bandt–Pompe)		
τ_{ch}	0.41	chaos threshold
τ_{st}	0.50	strong-coherence threshold
δ	4.6692016091	Feigenbaum constant
β (P1)	0.5	channel intensity
ψ	softplus	monotone map
ψ mode	level	or change / both
absolute-delta	True	form (8')
normalization	zscore	alternative rank / mad
P_{arm}	3	approach arm length
θ_e	1.25	level threshold
θ_{Δ}	0.40	change threshold
require change	True	(B γ)
θ_{res}	0.20	residual synergy; pre-specified via S0
$L_s, L_{s,\text{min}}$	3, 2	short scale
$L_m, L_{m,\text{min}}$	7, 4	medium scale
soft offset s_0	3.25	logistic offset
soft floor	0.25	$A_3^{\text{soft}} < \text{floor} \Rightarrow 0$
adaptive θ_e	on, $\kappa = 0.5$	baseline CV inflation
d_{min}	3 (default); 2 on bivariate Holter	bivariate CCTP
R_{max} (P2)	3	synthetic dual-channel
J (P2)	0	no jumps
η (P2)	0.35	synthetic dual-channel
$\alpha_{\text{Syn}}, \alpha_{\text{Surp}}$	0.6, 0.4	fixed

Appendix B — Relation to the existing nested formalism

Equation (2) of the 2026 framework [1] is already a linearly weighted integration of Φ_3 into ΔRECD . The present construction does not redefine Φ_3 ; it:

1. couples Level-3 structure to the legacy increment (1) used in applications;
2. conditions the weight on surplus evidence (not solely on $\lambda(\tau_s)$);
3. specifies activation, ablation, and validation so that the channel is scientifically auditable.

In unified notation, Proposal 1 is the special case

$$\Delta t_k = \Delta t_k^{\text{base}} g \alpha(1 + \beta A_3 \psi(\cdot)),$$

while equation (2) is

$$\Delta\text{RECD} = \sum_{\ell} \alpha_{\ell}(\lambda) \Phi_{\ell}^*.$$

A complete pipeline may report both indices: $T^{\text{legacy+P1}}$ for historical comparability and T^{nested} for mass fractions f_{ℓ} .

Appendix C — Scope boundaries

The following topics are outside the present analysis:

- axiomatic partial-information reformulation of Syn/Surp;
- claims about the ontology of physical time;
- grid-search optimization of β on outbreak or ventricular-fibrillation labels;
- replacement of the six-mode RQA consensus (complemented, not removed).

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Appendix D — Figures

S0 / S1 / S2-chaos — hardened A_3

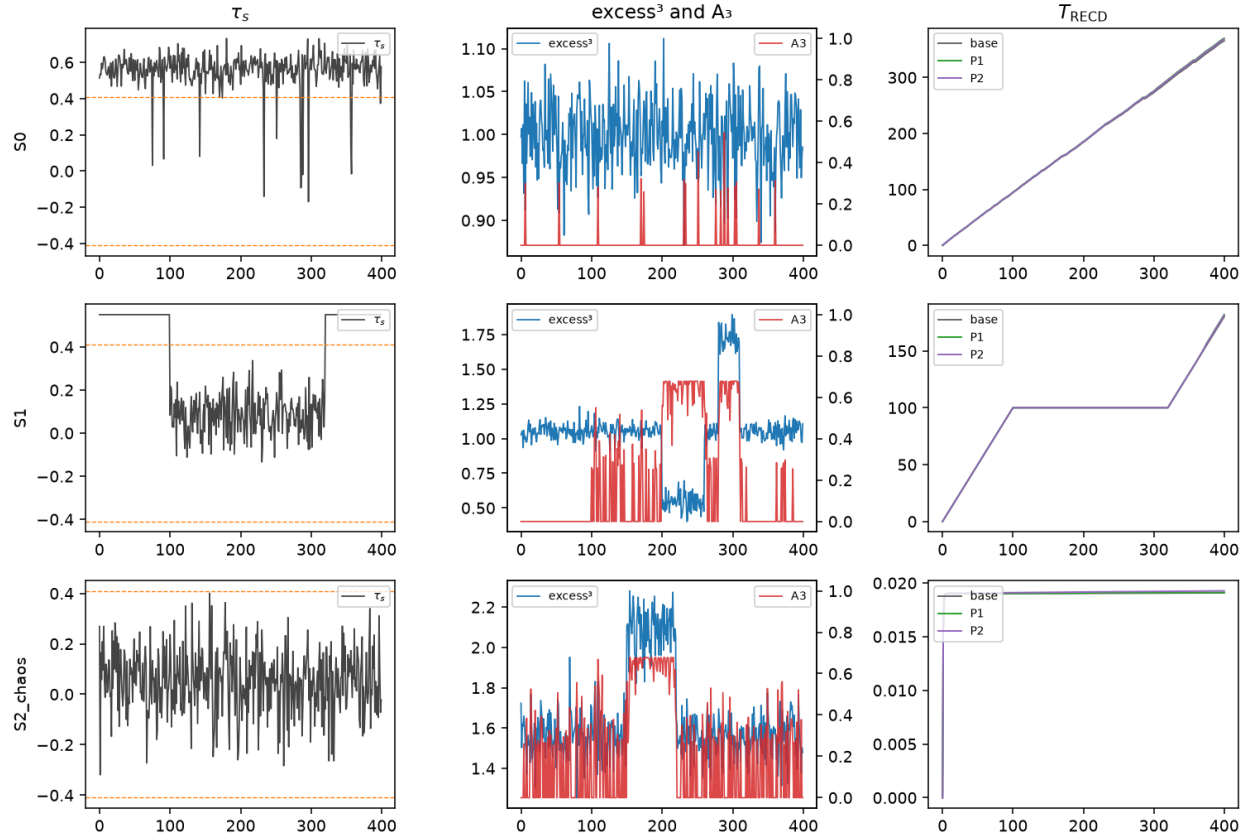


Figure 1: Synthetic S0-S2 overview

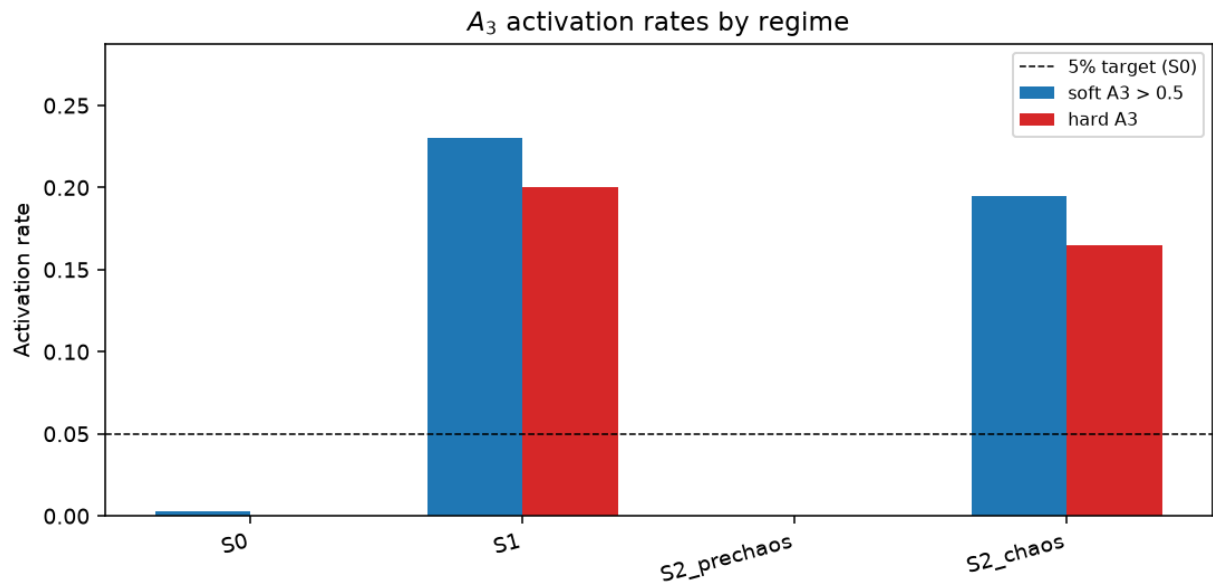


Figure 2: A_3 rates (synthetic)

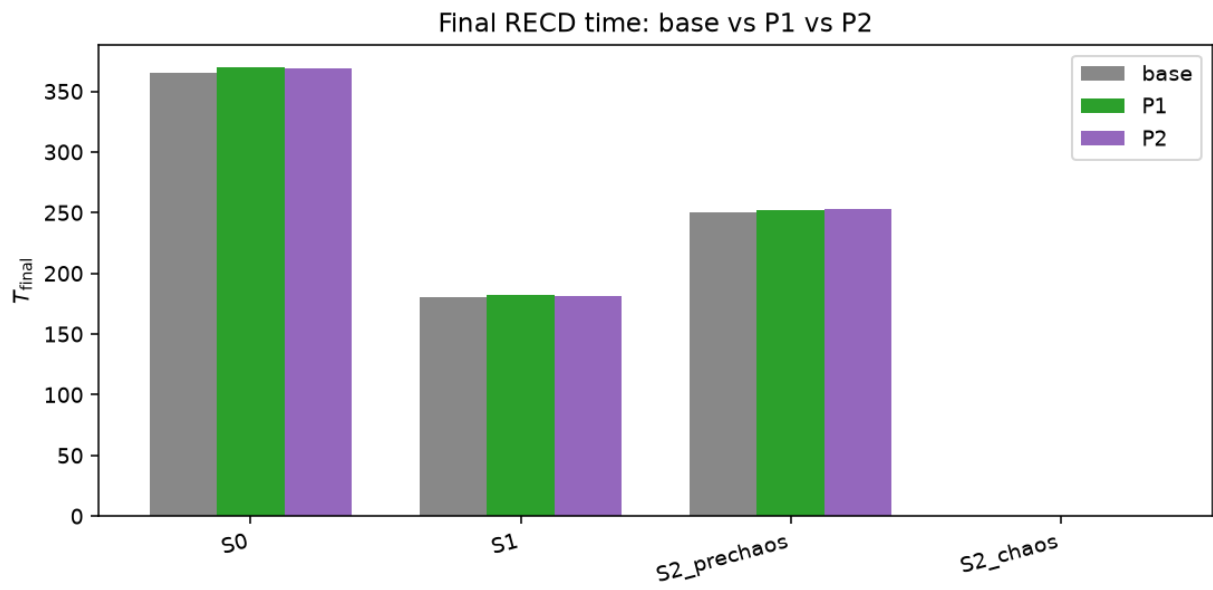


Figure 3: T_{final} (synthetic)

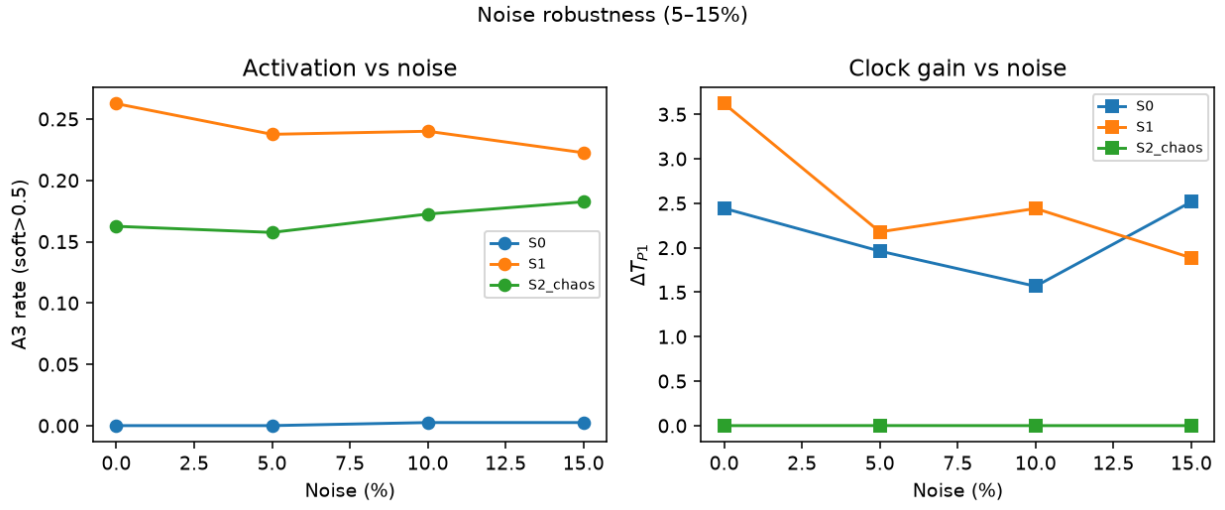


Figure 4: Noise robustness

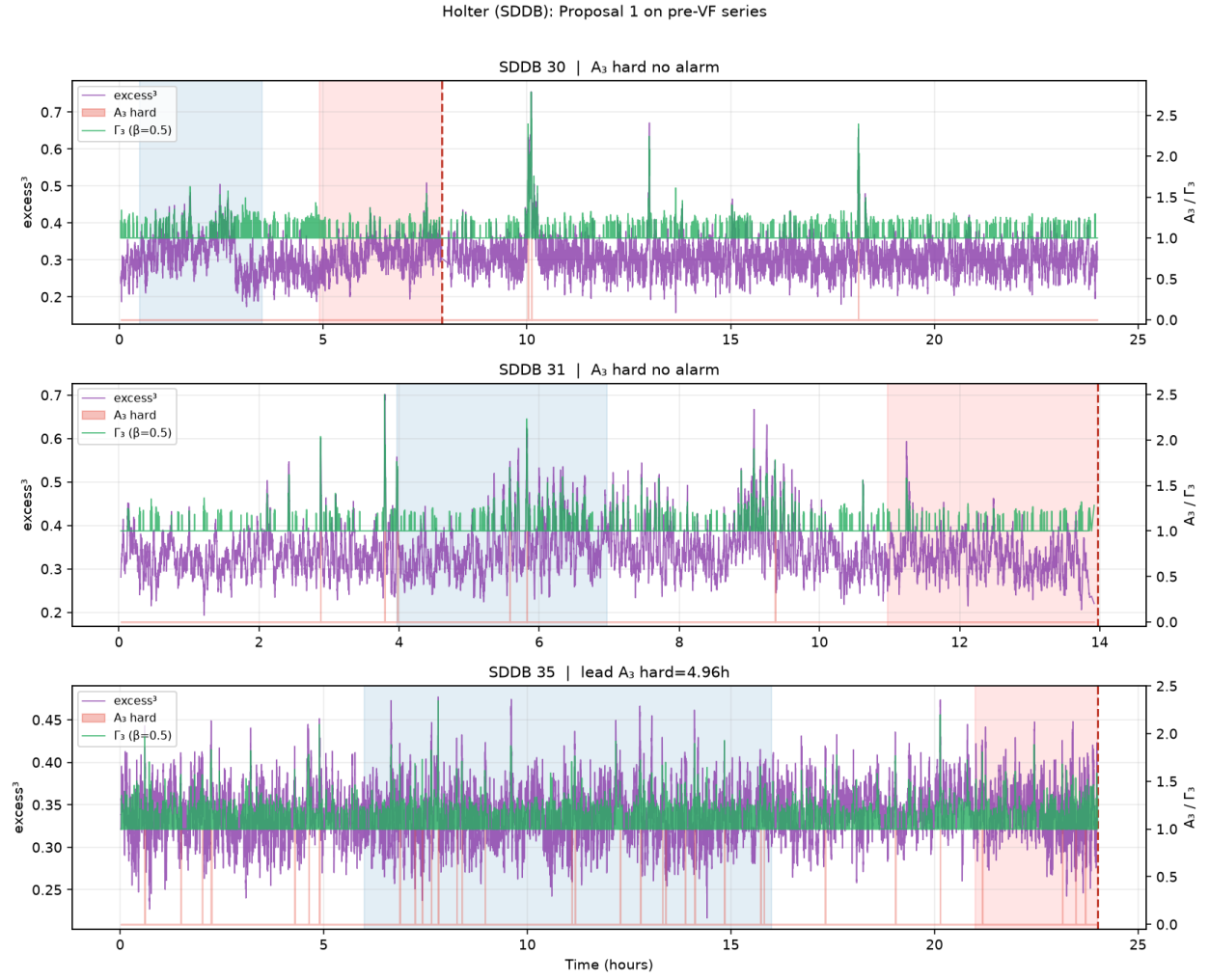


Figure 5: Holter (SDDB): pre-event series under Proposal 1 (Section 6.4)

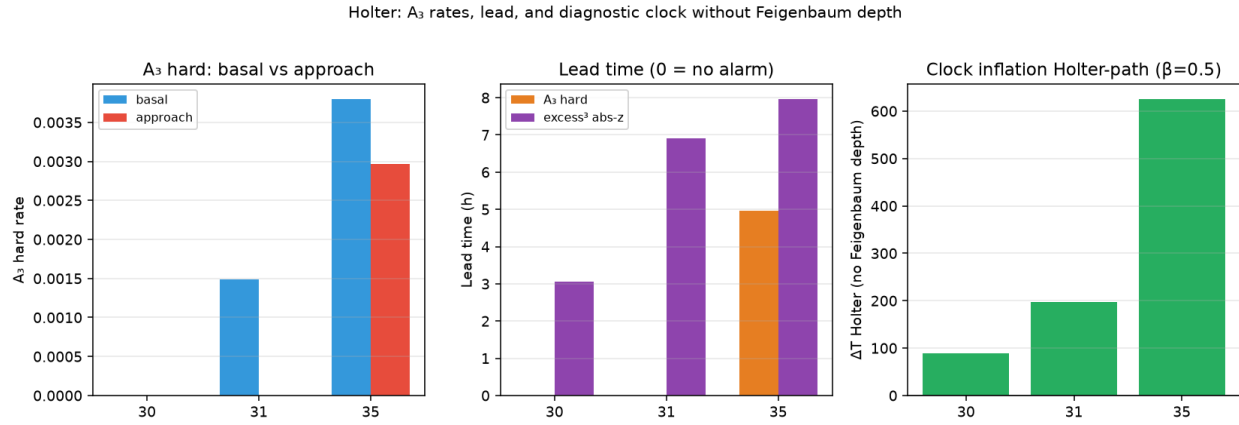


Figure 6: Holter: activation rates and lead (Section 6.4)

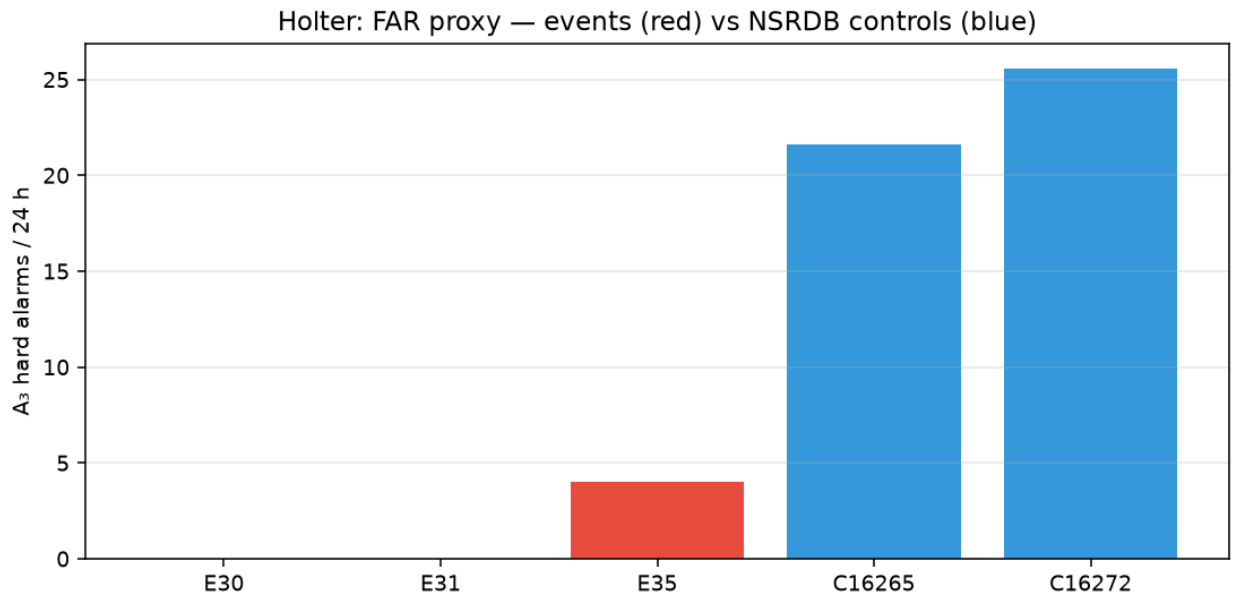


Figure 7: Holter: FAR proxy, events vs NSRDB controls (Section 6.4)

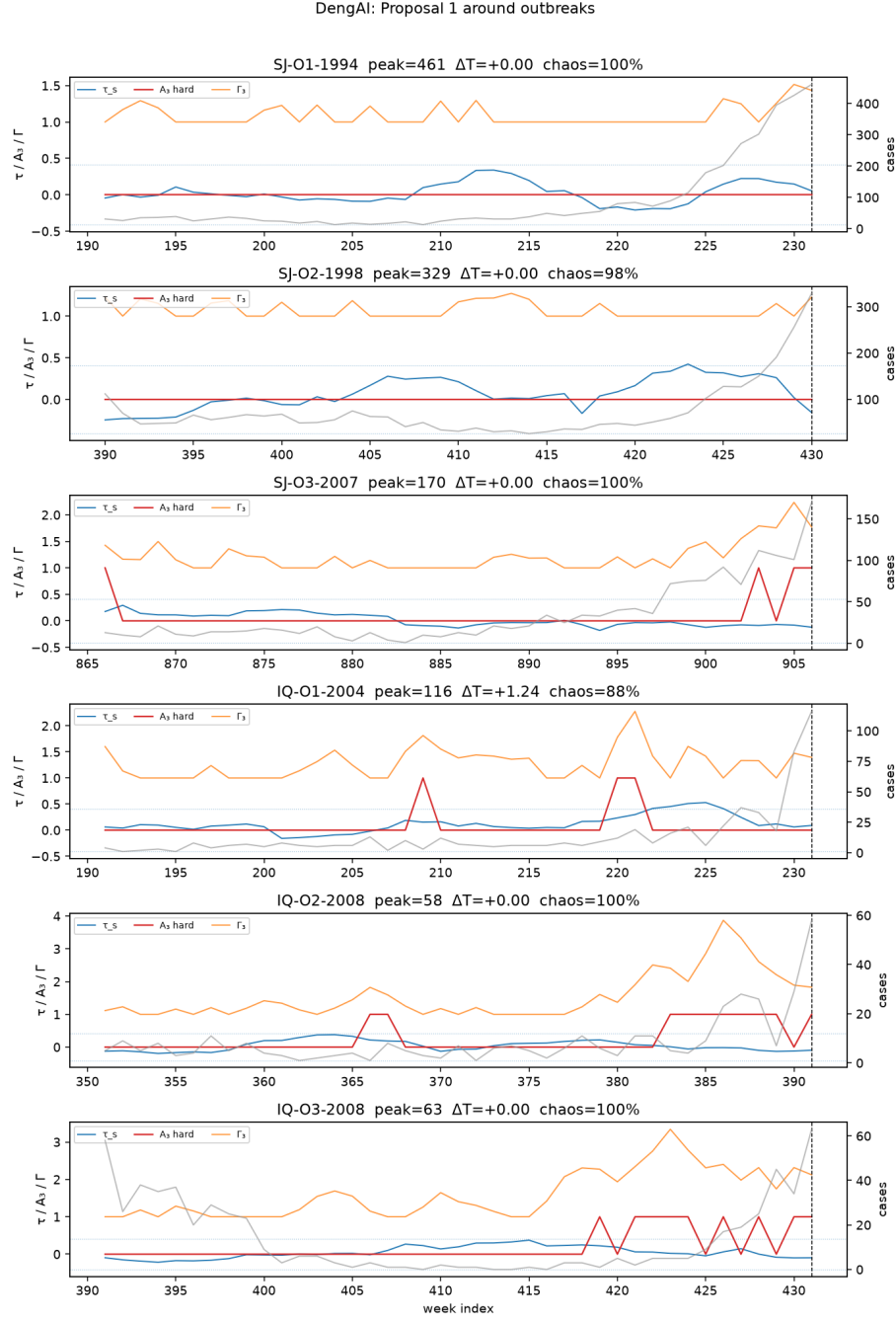


Figure 8: DengAI: pre-outbreak series under Proposal 1 (Section 6.4)

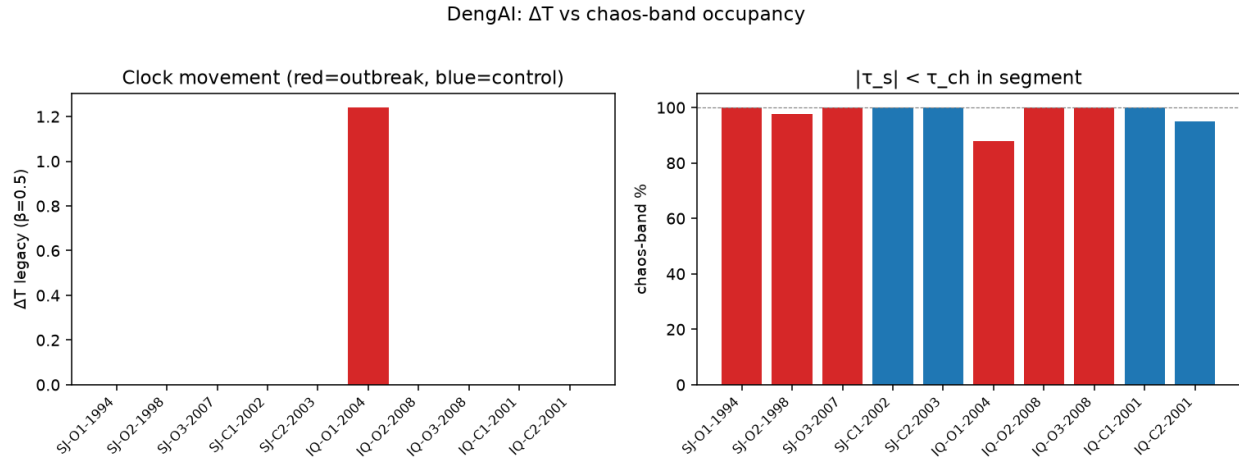


Figure 9: DengAI: ΔT vs chaos-band occupancy (Section 6.4)

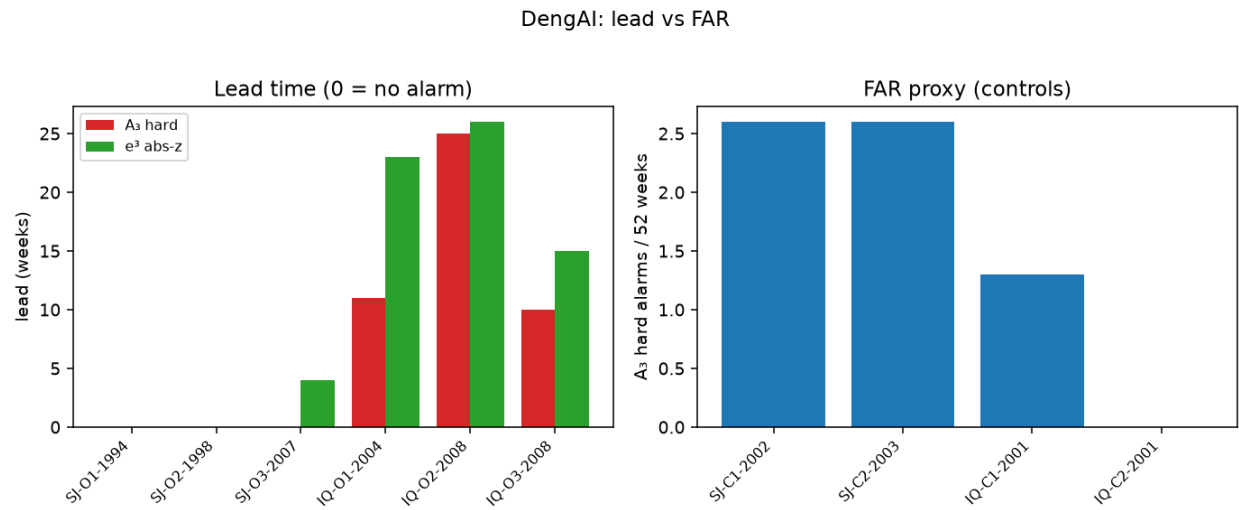


Figure 10: DengAI: lead vs FAR (Section 6.4)